

Zachary Huang

Los Angeles, California | [linkedin.com/in/zachuang2/](https://www.linkedin.com/in/zachuang2/) | (310) 213-0168 | zachhuang22@gmail.com

EDUCATION

Rice University

Houston, TX

ME in Bioengineering | Global Medical Innovation | GPA: 4.0

August 2025 - May 2026

University of California, San Diego

La Jolla, CA

B.S. in Mechanical Engineering with a specialization in Controls and Robotics | Major GPA: 3.96

Minor: General Biology | Honors: Cum Laude & Provost Honors

September 2021 - June 2025

EXPERIENCES

System Engineer | Relievio | Houston, TX

Aug 2025 - May 2026

- Collaborated with MD Anderson to address hospital-acquired pressure injuries (HAPI) affecting surgical patients during 3+ hour procedures, leading needs finding for a Class II wearable device targeting 50% HAPI reduction through real-time monitoring and active mitigation.
- Conducted 40+ hours of surgical observations documenting clinical workflows, led PUGH-matrix ideation, reducing 100+ concepts to 3 solutions, and created a design traceability matrix following ISO 13485 QMS.
- Designed and fabricated a wearable prototype with pressure monitoring and an active mitigation strategy that increases flow by 17% by validating through a silicone capillary membrane that measures different flow rates.

R&D Engineer | Texas Children's Hospital & BCM | Houston, TX

Aug 2025 - May 2026

- Addressed the diagnostic gap affecting 190,000+ pediatric patients annually who lack objective breathing quantification tools, executing rapid prototyping development for a respiratory monitoring device.
- Conducted literature review and clinical workflow to define user needs and regulatory pathway aligned with ISO 13485 QMS, designed a prototype to track diaphragm expansion to translate respiratory rate measurement.
- Achieved 97% respiratory rate accuracy and peak detection validated against manual monitoring and improved device wearability by fabricating a light-weight 80g device.

Product Research & Development Intern | Procter & Gamble | Cincinnati, OH

May 2025 - Aug 2025

- Led new product development for next-generation Class II feminine hygiene product from early innovation, addressing consumer needs for improved comfort and performance while reducing form factor by 30%.
- Conducted 12 interviews with two primary need focuses on disruptive designs, characterization testing (tensile, SEM) across 10 materials, and designed verification protocols based on absorbency, strength, and biocompatibility.
- Executed 50% radial expansion performance improvement through testing, and authored technical documentation on fabrication to achieve the specific blended material that maximizes the expansion.

Technical Mechanical Leader | UC San Diego MAE | La Jolla, CA

Sept 2024 - Apr 2025

- Guided 5-person team through Class III endovascular device for blood clotting removal for diabetic patients from concept through verification testing, design controls, and DHF maintenance within QMS framework.
- Acted as subject matter expert on vascular biomaterials, conducting literature review and executing 15+ feasibility studies, achieving 90% clot engagement with less than 2mm fragmentation.
- Engineered modular gearbox system (300-500 rpm, 15:1 ratio) with GD&T specifications, creating a design package (15+ detail drawings), reducing the catheter profile 60% (5mm to 2mm).

Process Engineering Intern | Procter & Gamble Tambrands | Auburn, ME

June 2024 - Sept 2024

- Designed automated material handling system, conducted 100+ hour study validating system performance, and collaborated with cross-functional teams (quality, manufacturing, process) to track implementation strategies.
- Achieved 93% system uptime delivering \$250K/year projected cost savings, proposed 50% capacity expansion (4 to 8 automated vehicles) validated through performance data, and maintained QMS compliance.

SKILLS

CAD Software: SolidWorks, Autodesk Inventor, AutoCAD, 3D Printing (FDM, SLA), GD&T Application & Engineering Drawings, Design for Manufacturing (DFM)

Clinical & Domain Knowledge: Surgical OR Workflow Observation, Device Human Factors, Pre-clinical Testing

Medical Device Development: FDA Design Controls & Testing, JAMA, Control System Design

Quality & Regulatory: ISO 14971-1:2019, ISO 10993-1:2018, ISO 13485:2016, and ISO 286

Documentation: Design Specs, Verification & Test Protocols, DHF/technical documents, Invention Disclosures